

1. Classification of Robots

Robots can be classified based on **structure, motion, control, and application**.

1.1 Based on Mechanical Structure (Configuration)

1. **Cartesian Robot (Rectangular / Gantry Robot)**
 - Motion along X, Y, Z axes
 - Uses linear joints
 - High accuracy and rigidity
 - **Applications:** CNC machines, pick-and-place, 3D printing
2. **Cylindrical Robot**
 - Cylindrical coordinate system (rotational + linear)
 - One rotary joint and two linear joints
 - **Applications:** Material handling, assembly operations
3. **Spherical (Polar) Robot**
 - Two rotary joints + one linear joint
 - Large work envelope
 - **Applications:** Welding, casting, machine loading
4. **SCARA Robot (Selective Compliance Assembly Robot Arm)**
 - Rigid in vertical direction, flexible in horizontal
 - High-speed assembly
 - **Applications:** Electronics assembly, packaging
5. **Articulated Robot**
 - Multiple rotary joints (similar to human arm)
 - Typically 4 to 6 DOF
 - **Applications:** Welding, painting, material handling

❖ Cartesian Robot (Rectangular / Gantry Robot)

1. Introduction

A **Cartesian Robot**, also known as a **Rectangular Robot** or **Gantry Robot**, is a type of industrial robot whose motion is controlled along **three mutually perpendicular linear axes (X, Y, and Z)**. It operates in a **Cartesian coordinate system**, where each axis corresponds to a linear movement.

2. Construction of Cartesian Robot

A Cartesian robot consists of the following main components:

2.1 Mechanical Structure

- Three linear slides arranged orthogonally
- Each slide represents one axis:
 - **X-axis** – Horizontal movement

- **Y-axis** – Horizontal or vertical movement (depending on design)
- **Z-axis** – Vertical movement (up and down)

2.2 Actuators

- Linear actuators
- DC motors, stepper motors, or servo motors
- Ball screws, lead screws, belt drives, or rack-and-pinion mechanisms

2.3 End Effector

- Mounted at the end of Z-axis
- Examples:
 - Grippers
 - Vacuum cups
 - Welding torches
 - Cutting tools

2.4 Control System

- PLC or microcontroller
 - Motion controller
 - Motor drivers
 - Feedback devices (encoders, limit switches)
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3. Working Principle

- Motion is achieved by **independent control of linear axes**.
 - Each axis moves in a straight line.
 - The position of the end effector is defined by **(X, Y, Z) coordinates**.
 - Simultaneous motion of axes allows movement to any point within the work envelope.
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4. Degrees of Freedom (DOF)

- Each linear axis contributes **one DOF**.
- Typical Cartesian robot has:
 - **3 DOF** – Position control
 - Additional DOF may be added using a wrist or rotary joints for orientation

5. Coordinate System

- Uses **rectangular coordinate system**.
 - Position is specified by three parameters:
 - X → length
 - Y → width
 - Z → height
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6. Types of Cartesian Robots

6.1 Table-Top Cartesian Robot

- Compact design
- Used in small assembly tasks

6.2 Cantilever Cartesian Robot

- Supported on one side
- Medium load handling

6.3 Gantry Cartesian Robot

- Large structure supported on both sides
 - High load capacity
 - Covers large working area
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7. Advantages of Cartesian Robot

1. Simple mechanical design
 2. High positional accuracy and repeatability
 3. Easy programming and control
 4. High load carrying capacity
 5. Rigid structure
 6. Easy maintenance
 7. Modular design (easy expansion)
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8. Disadvantages of Cartesian Robot

1. Large space requirement
 2. Limited flexibility compared to articulated robots
 3. Bulky structure
 4. Less suitable for complex orientations
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9. Applications of Cartesian Robots

9.1 Industrial Applications

- Pick-and-place operations
- CNC machine loading/unloading
- 3D printing
- Packaging
- Palletizing

❖ Cylindrical Robot

1. Introduction

A **Cylindrical Robot** is an industrial robot whose motion is defined in a **cylindrical coordinate system**.

It consists of **one rotary joint and at least two linear (prismatic) joints**, allowing the robot to move in **circular and vertical directions**.

The working area of a cylindrical robot is **cylindrical in shape**.

2. Construction of Cylindrical Robot

The main components of a cylindrical robot are:

2.1 Base

- Fixed or rotating base
- Provides **rotational motion** around the vertical axis

2.2 Vertical Column

- Mounted on the base
- Supports the arm structure

2.3 Arm (Radial Arm)

- Extends and retracts horizontally
- Provides **linear motion (radial movement)**

2.4 Vertical Slide

- Moves the arm **up and down**
- Provides vertical linear motion

2.5 Actuators

- DC motors
- Stepper motors
- Servo motors

2.6 End Effector

- Gripper, welding torch, suction cup, etc.
 - Mounted at the end of the arm
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3. Joint Configuration

A typical cylindrical robot has the following joints:

Joint Type	Motion
Revolute Joint (R)	Base rotation
Prismatic Joint (P)	Vertical movement
Prismatic Joint (P)	Radial (in-out) movement

Joint notation: R-P-P

4. Working Principle

1. The **base rotates** about the vertical axis (θ motion).
 2. The **arm extends or retracts** radially.
 3. The **arm moves vertically** along the column.
 4. Combination of these three motions positions the end effector at any point within the cylindrical work envelope.
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5. Degrees of Freedom (DOF)

- **3 DOF** for position control:
 - 1 rotational
 - 2 linear
 - Additional DOF may be added using wrist joints for orientation.
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6. Coordinate System

The position of the end effector is defined using:

- **r** → radial distance
- **θ** → angular position
- **z** → vertical position

This forms a **cylindrical coordinate system**.

7. Work Envelope

- The work envelope is **cylindrical or hollow cylindrical**.
- Determined by:
 - Maximum and minimum arm extension
 - Range of base rotation
 - Vertical stroke length

8. Types of Cylindrical Robots

8.1 Standard Cylindrical Robot

- Basic R-P-P configuration
- Used for general material handling

8.2 Telescopic Cylindrical Robot

- Uses telescopic arms
 - Extended reach
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9. Advantages of Cylindrical Robots

1. Simple mechanical structure
 2. Good vertical reach
 3. Can handle moderate payloads
 4. Easier control than articulated robots
 5. Occupies less floor space than Cartesian robots
 6. Suitable for machine loading/unloading
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10. Disadvantages of Cylindrical Robots

1. Limited dexterity
2. Lower flexibility than articulated robots
3. Accuracy affected by arm extension
4. Not suitable for complex orientations

11. Applications of Cylindrical Robots

11.1 Industrial Applications

- Machine tool loading and unloading
- Die casting
- Spot welding
- Material handling
- Assembly operations

11.2 Other Applications

- Foundry work
 - Palletizing
 - Automated storage systems
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❖ Spherical (Polar) Robot

1. Introduction

A **Spherical Robot**, also known as a **Polar Robot**, is an industrial robot whose motion is defined in a **spherical (polar) coordinate system**.

The robot's arm is capable of **rotational movement in two directions and linear movement in one direction**, allowing the end effector to reach points within a **spherical-shaped work envelope**.

This robot resembles a **human shoulder movement**, providing wide reach and flexibility.

2. Construction of Spherical (Polar) Robot

The main components of a spherical robot are:

2.1 Base

- Fixed base mounted on the floor
- Provides **rotational movement** about the vertical axis

2.2 Arm Assembly

- Consists of two main arms:
 - **Arm rotation joint** (shoulder)
 - **Arm elevation joint** (up/down angular movement)

2.3 Telescopic Arm

- Extends and retracts linearly
- Controls radial distance of the end effector

2.4 Actuators

- DC motors
- Servo motors
- Hydraulic or pneumatic actuators (for heavy loads)

2.5 End Effector

- Gripper, spray gun, welding torch, etc.
- Mounted at the arm tip

3. Joint Configuration

A typical spherical robot uses the following joint arrangement:

Joint Type	Motion
Revolute Joint (R)	Base rotation
Revolute Joint (R)	Arm elevation

Joint Type	Motion
Prismatic Joint (P)	Arm extension

Joint notation: R-R-P

4. Working Principle

1. The **base rotates** about the vertical axis (θ_1).
 2. The **arm rotates up or down** about a horizontal axis (θ_2).
 3. The **telescopic arm extends or retracts** linearly (r).
 4. Combination of these motions positions the end effector anywhere within the spherical workspace.
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5. Degrees of Freedom (DOF)

- **3 DOF** for positioning:
 - 2 rotational
 - 1 linear
- Additional DOF can be added using wrist joints for orientation control.

6. Coordinate System

The position of the end effector is defined by:

- $r \rightarrow$ radial distance
- $\theta \rightarrow$ base rotation angle
- $\phi \rightarrow$ elevation angle

This forms a **spherical (polar) coordinate system**.

7. Work Envelope

- The work envelope is **spherical or partial spherical**.
- Influenced by:
 - Maximum arm extension
 - Rotation limits of joints
 - Elevation angle limits

8. Types of Spherical Robots

8.1 Standard Polar Robot

- Basic R-R-P configuration
- Used for general industrial operations

8.2 Telescopic Polar Robot

- Equipped with multi-stage telescopic arms
- Larger reach

9. Advantages of Spherical (Polar) Robots

1. Large working envelope
 2. Good flexibility
 3. Requires less floor space
 4. Can reach over and around obstacles
 5. Suitable for heavy load handling
 6. Simple kinematic structure compared to articulated robots
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10. Disadvantages of Spherical (Polar) Robots

1. Lower accuracy compared to Cartesian robots
 2. Complex control compared to cylindrical robots
 3. Mechanical complexity due to rotating joints
 4. Maintenance can be difficult
 5. Limited speed in some operations
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11. Applications of Spherical Robots

11.1 Industrial Applications

- Die casting
- Spot welding
- Arc welding
- Machine loading and unloading
- Foundry operations

11.2 Other Applications

- Spray painting
 - Material transfer
 - Handling hazardous materials
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❖ SCARA Robot (Selective Compliance Assembly Robot Arm)

1. Introduction

A **SCARA Robot** stands for **Selective Compliance Assembly Robot Arm**.

It is a special type of industrial robot designed mainly for **high-speed, high-precision assembly operations**, especially in the **horizontal plane**.

The term “**Selective Compliance**” means:

- **Compliant (flexible)** in the **X-Y plane**
- **Rigid** in the **Z-direction**

This unique feature makes SCARA robots ideal for **insertion, pick-and-place, and small part assembly tasks**.

2. Construction of SCARA Robot

The main parts of a SCARA robot are:

2.1 Base

- Fixed base mounted on the floor or table
- Houses the first rotary motor

2.2 First Arm (Link-1)

- Rotates around the vertical axis
- Provides horizontal movement

2.3 Second Arm (Link-2)

- Connected to the first arm
- Rotates in the same horizontal plane

2.4 Vertical Slide (Z-axis)

- Provides up-and-down linear motion
- Ensures rigid vertical movement

2.5 Wrist (Optional)

- Provides rotation of end effector around Z-axis

2.6 End Effector

- Gripper, vacuum cup, screwdriver, etc.
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3. Joint Configuration

A typical SCARA robot uses the following joint arrangement:

Joint	Type	Motion
Joint-1	Revolute (R)	Rotation in X-Y plane
Joint-2	Revolute (R)	Rotation in X-Y plane
Joint-3	Prismatic (P)	Vertical (Z-axis) movement
Joint-4 (optional)	Revolute (R)	Wrist rotation

4. Working Principle

1. The **first and second arms rotate** in the horizontal plane.
 2. Combined rotation allows the end effector to reach any point in the X-Y plane.
 3. The **Z-axis slide** moves the arm up and down.
 4. The wrist rotation (if present) adjusts the orientation of the tool.
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5. Degrees of Freedom (DOF)

- **4 DOF (commonly):**
 - 2 rotational (X-Y plane)
 - 1 linear (Z-axis)
 - 1 rotational (end effector orientation)
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6. Coordinate System

- Uses a **cylindrical coordinate system**
 - Coordinates:
 - $\theta_1, \theta_2 \rightarrow$ rotational angles
 - **Z** $\rightarrow$ vertical movement
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7. Work Envelope

- Work envelope is **donut-shaped (annular)** in the horizontal plane
 - Vertical motion adds height
 - Suitable for table-top operations
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8. Selective Compliance Explained

- **Compliant in X-Y plane**
 - Allows slight flexibility
 - Helps in easy part insertion
- **Rigid in Z-axis**
 - Ensures accurate vertical placement
 - Prevents bending during insertion

This feature reduces damage to parts and improves assembly speed.

9. Advantages of SCARA Robot

1. High speed and acceleration
 2. High positional accuracy and repeatability
 3. Compact size
 4. Excellent for repetitive tasks
 5. Simple programming
 6. Lower cost compared to articulated robots
 7. High productivity
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10. Disadvantages of SCARA Robot

1. Limited vertical reach
2. Restricted to assembly-type operations
3. Limited flexibility compared to 6-axis robots
4. Smaller working envelope

11. Applications of SCARA Robots

11.1 Industrial Applications

- Electronic component assembly
- PCB assembly
- Pick-and-place operations
- Screw driving
- Packaging

11.2 Other Applications

- Pharmaceutical assembly
 - Medical device manufacturing
 - Small part handling
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❖ Articulated Robot

1. Introduction

An **Articulated Robot** is a type of industrial robot that consists of **rotary (revolute) joints only**, arranged in a chain similar to a **human arm**.

Because of this structure, articulated robots are also known as **jointed-arm robots**.

These robots are the **most widely used industrial robots** due to their **high flexibility, wide working range, and ability to perform complex tasks**.

2. Construction of Articulated Robot

The main components of an articulated robot are:

2.1 Base

- Fixed to the floor or mounted on a platform
- Allows rotation of the entire arm about the vertical axis

2.2 Arm (Manipulator)

The arm is divided into sections:

1. **Shoulder Joint**
 - Rotates the arm forward and backward
2. **Elbow Joint**
 - Bends and extends the arm
3. **Wrist Assembly**
 - Consists of multiple rotary joints
 - Provides orientation control (roll, pitch, yaw)

2.3 Links

- Rigid members connecting joints
- Mimic upper arm, forearm, and wrist

2.4 Actuators

- Servo motors (commonly used)
- Hydraulic actuators (for heavy-duty robots)

2.5 End Effector

- Gripper, welding torch, spray gun, cutting tool, etc.

3. Joint Configuration

A typical articulated robot has **4 to 6 rotary joints**.

Common Joint Arrangement

- **Joint-1:** Base rotation
- **Joint-2:** Shoulder rotation
- **Joint-3:** Elbow rotation
- **Joint-4:** Wrist pitch
- **Joint-5:** Wrist yaw
- **Joint-6:** Wrist roll

Joint notation: R-R-R-R-R-R

4. Working Principle

1. The base rotates to position the robot.

2. Shoulder and elbow joints move the arm forward/backward and up/down.
 3. Wrist joints orient the end effector in any direction.
 4. Coordinated motion of all joints allows the robot to reach and orient the tool at any desired point.
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5. Degrees of Freedom (DOF)

- **Minimum:** 4 DOF
- **Standard industrial robot:** 6 DOF

Why 6 DOF?

- 3 DOF for position (X, Y, Z)
 - 3 DOF for orientation (roll, pitch, yaw)
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6. Coordinate System

- Uses a **joint-based (angular) coordinate system**
 - Each joint angle defines the position and orientation of the end effector
 - Kinematics is more complex compared to Cartesian robots
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7. Work Envelope

- The work envelope is **irregular and spherical-like**
 - Very large reach relative to its footprint
 - Can reach around obstacles
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8. Types of Articulated Robots

8.1 4-Axis Articulated Robot

- Used for simple pick-and-place operations

8.2 5-Axis Articulated Robot

- Improved orientation control

8.3 6-Axis Articulated Robot

- Most common
 - Full position and orientation control
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9. Advantages of Articulated Robots

1. High flexibility and dexterity
 2. Can perform complex and multi-directional tasks
 3. Large working envelope
 4. Can reach around obstacles
 5. Suitable for many industrial applications
 6. High payload capacity (especially hydraulic robots)
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10. Disadvantages of Articulated Robots

1. Complex mechanical structure
 2. Complicated control and programming
 3. Lower accuracy compared to Cartesian robots
 4. Higher cost
 5. Requires skilled maintenance
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11. Applications of Articulated Robots

11.1 Industrial Applications

- Arc and spot welding
- Spray painting
- Material handling
- Assembly operations
- Machining and cutting

11.2 Other Applications

- Medical surgery
- Space exploration
- Research and education
- Humanoid robots

1.2 Based on Control Method

1. **Non-Servo Robots**
 - Open-loop control
 - Limited accuracy
 2. **Servo Robots**
 - Closed-loop control
 - Feedback from sensors
 - High precision and accuracy
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1.3 Based on Application

- Industrial robots
 - Medical robots
 - Service robots
 - Mobile robots
 - Humanoid robots
 - Military and space robots
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2. Links and Joints

2.1 Links

- **Link** is a rigid body that connects two joints.
 - Forms the structure of the robot.
 - **Types of links:**
 - Rigid links
 - Flexible links
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2.2 Joints

A **joint** allows relative motion between two links.

Types of Joints

1. **Revolute Joint (R)**
 - Rotational motion
 - Example: Elbow joint
2. **Prismatic Joint (P)**
 - Linear motion
 - Example: Sliding mechanism
3. **Spherical Joint**
 - Rotation in three axes
4. **Cylindrical Joint**
 - Combination of linear + rotary motion
5. **Helical Joint**
 - Screw-type motion

3. Degree of Freedom (DOF)

- **Degree of Freedom** is the number of independent motions a robot can perform.
- Each joint contributes **one DOF**.

Examples

- 2 joints → 2 DOF
- 6-axis industrial robot → 6 DOF

Importance of DOF

- Determines robot flexibility
 - Higher DOF → more complex tasks
 - Minimum **6 DOF** required for full spatial manipulation (position + orientation)
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4. Motors Used in Robotics

Motors convert **electrical energy into mechanical motion**.

4.1 DC Motors

Working Principle

- Operates on Fleming's Left Hand Rule
- Rotation due to interaction of magnetic field and current

Types

- Brushed DC motor
- Brushless DC motor (BLDC)

Advantages

- Simple control
- High torque
- Easy speed control

Disadvantages

- Brushes wear out (brushed motor)

Applications

- Mobile robots
- Robotic arms
- Line follower robots

4.2 Stepper Motors

Working Principle

- Moves in discrete steps
- Controlled by pulse signals

Types

- Permanent magnet
- Variable reluctance
- Hybrid stepper motor

Advantages

- Precise position control
- No feedback required

Disadvantages

- Low efficiency
- Resonance issues

Applications

- CNC machines
- 3D printers
- Robotic joints

4.3 Servo Motors

Construction

- DC motor
- Gear system
- Position sensor (potentiometer)
- Control circuit

Working

- Closed-loop control
- Controlled using PWM signal

Advantages

- High accuracy
- Precise position control
- High torque

Disadvantages

- Costly
- Limited rotation (usually 0–180°)

Applications

- Robotic arms
 - Humanoid robots
 - Camera gimbals
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5. Power Transmission – Types of Gears

Gears transmit power and motion between shafts.

5.1 Spur Gear

- Teeth parallel to shaft
 - Simple and efficient
 - **Used in:** Low-speed applications
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5.2 Helical Gear

- Teeth inclined to shaft
 - Smooth and quiet operation
 - **Used in:** High-speed robots
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5.3 Bevel Gear

- Used to change direction of rotation
 - Shafts intersect at an angle
 - **Used in:** Robotic joints
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5.4 Worm Gear

- High reduction ratio
 - Self-locking
 - **Used in:** Robotic arms, elevators
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5.5 Planetary (Epicyclic) Gear

- Compact size
 - High torque
 - **Used in:** Servo motors
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6. Robotic Sensors

Sensors provide **feedback** to robots.

6.1 Position Sensors

- Potentiometers
 - Encoders (incremental, absolute)
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6.2 Proximity Sensors

- Infrared (IR)
 - Ultrasonic
 - Capacitive
 - Inductive
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6.3 Force and Torque Sensors

- Measure applied force
 - Used in assembly and gripping
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6.4 Vision Sensors

- Cameras
 - Image processing systems
 - Used for object recognition
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6.5 Environmental Sensors

- Temperature
 - Gas
 - Humidity
 - Light sensors
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7. Applications of Robots

7.1 Industrial Applications

- Welding
 - Painting
 - Assembly
 - Material handling
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7.2 Medical Applications

- Surgery (Da Vinci robot)
 - Rehabilitation
 - Prosthetics
-

7.3 Military Applications

- Bomb disposal
 - Surveillance
 - Unmanned vehicles
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7.4 Space Exploration

- Mars rovers
 - Satellite servicing
-

7.5 Service Applications

- Cleaning robots
 - Delivery robots
 - Domestic robots
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8. Software Used for Robot Programming

8.1 Programming Languages

- C / C++
 - Python
 - Java
 - MATLAB
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8.2 Robot Operating System (ROS)

- Open-source framework
 - Libraries for navigation, perception, control
 - Widely used in research and industry
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8.3 Simulation Software

- Gazebo
 - Webots
 - V-REP (CoppeliaSim)
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8.4 Industrial Robot Programming Software

- ABB – RAPID
 - FANUC – KAREL
 - KUKA – KRL
 - Universal Robots – Polyscope
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8.5 Embedded Platforms

- Arduino IDE
- MPLAB
- Keil
- PlatformIO

Links and Joints in Robotics

In robotics, a robot manipulator (robot arm) is made up of **rigid parts connected together to perform movement and tasks**. These parts are mainly classified as **Links** and **Joints**. They form the mechanical structure of a robot and determine its **motion and flexibility**.

1. Links in Robotics

Definition

A **link** is a **rigid structural component of a robot** that connects two joints. Links act as the **bones of a robot manipulator** and provide support and structure.

Characteristics of Links

- They are **rigid bodies**.
- They connect **two or more joints**.
- They transmit **motion and force** from one joint to another.
- They form the **mechanical arm of the robot**.

Types of Links

1. Rigid Links

- The most common type of links.
- They do not change shape during movement.
- Used in industrial robots.

Example: Metal arms in robot manipulators.

2. Flexible Links

- These links may **bend or deform slightly** during motion.
- Used in **lightweight robots**.

Example: Robotic arms used in space applications.

3. Fluid Links

- Motion is transmitted through **fluids such as hydraulic systems**.
- Used in heavy machinery robots.

Functions of Links

- Connect robot joints.
- Support mechanical structure.
- Transfer motion between joints.
- Carry sensors, actuators, and end-effectors.

Example of Robot Links

Base ---- Link1 ---- Link2 ---- Link3 ---- End Effector

Each **link connects two joints** and contributes to the robot's movement.

2. Joints in Robotics

Definition

A **joint** is a **mechanical connection between two links** that allows motion between them.

Joints determine the **type and direction of movement** in a robot.

Characteristics of Joints

- Provide **relative motion between links**.
 - Define the **degrees of freedom (DOF)** of a robot.
 - Can allow **rotational or translational movement**.
-

Types of Joints in Robotics

1. Revolute Joint (R Joint)

- Allows **rotational motion** around a fixed axis.
- Similar to a **hinge joint**.

Example

- Elbow joint of a human arm.

Motion

Rotation



Link1 ----- Joint ----- Link2

Applications

- Robotic arms
 - Industrial manipulators
-

2. Prismatic Joint (P Joint)

- Allows **linear or sliding motion**.
- One link moves **along the axis of another link**.

Example

- Hydraulic pistons.

Motion

Sliding Motion

Link1 =====> Link2

Applications

- CNC machines
 - Linear robotic actuators
-

3. Cylindrical Joint (C Joint)

- Allows **two types of motion**:
 - Rotation
 - Linear sliding

Motion

- Combination of **revolute + prismatic joint**.

Applications

- Robotic assembly systems.
-

4. Spherical Joint (S Joint)

- Allows **rotation in multiple directions**.
- Similar to a **ball-and-socket joint**.

Example

- Human shoulder joint.

Applications

- Humanoid robots
 - Robotic arms requiring flexible movement.
-

5. Planar Joint (E Joint)

- Allows motion in a **plane**.
- Provides **two translational movements and one rotational movement**.

Applications

- Mobile robot platforms.
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Relationship Between Links and Joints

Links and joints together form a **robot manipulator structure**.

Base

|

Joint1

|

Link1

|

Joint2

|

Link2

|

Joint3

|

End Effector

- **Links** provide structure.
- **Joints** provide motion.

Degree of Freedom (DOF)

The **degree of freedom** is the **number of independent movements a robot can perform**.

Example:

Joint Type DOF

Revolute 1

Prismatic 1

Spherical 3

More joints → **greater flexibility and movement capability**.

Applications of Links and Joints

- Industrial robotic manipulators
- Welding robots
- Assembly robots
- Medical surgical robots
- Space exploration robots

Motors in Robotics

Motors are essential components in robotics that **convert electrical energy into mechanical motion**. They are used to move robot joints, wheels, arms, and other mechanical parts. The most commonly used motors in robotics are **DC Motors, Stepper Motors, and Servo Motors**.

1. DC Motor in Robotics

Definition

A **DC (Direct Current) motor** is an electrical device that converts **direct current electrical energy into rotational mechanical motion**.

Working Principle

DC motors work on the principle of **electromagnetism**.

When electric current flows through a conductor placed in a magnetic field, a **force is produced**, causing the rotor to rotate.

Main Components

- **Stator** – Provides magnetic field
- **Rotor (Armature)** – Rotating part of the motor
- **Commutator** – Reverses current direction
- **Brushes** – Supply current to the armature

Working Process

1. Electric current flows through the armature.
2. The magnetic field interacts with the current.
3. A force is generated causing rotation.
4. The commutator maintains continuous rotation.

Advantages

- Simple design
- Low cost
- Easy speed control
- High starting torque

Disadvantages

- Brushes wear out
- Requires maintenance
- Less precise position control

Applications in Robotics

- Robot wheels

- Conveyor robots
 - Mobile robots
 - Low-cost robotic systems
-

2. Stepper Motor in Robotics

Definition

A **stepper motor** is a motor that moves in **discrete steps instead of continuous rotation**. Each electrical pulse moves the motor by a fixed angle.

Working Principle

The motor rotates step-by-step when **electrical pulses** are applied to its windings.

Structure

Main parts include:

- **Stator** (electromagnets)
- **Rotor** (permanent magnet or soft iron)

Step Angle

The **step angle** determines how much the motor rotates per pulse.

Example:

If a motor has **200 steps per revolution**:

$$\text{Step Angle} = \frac{360}{200} = 1.8^\circ$$

Types of Stepper Motors

1. **Permanent Magnet Stepper Motor**
2. **Variable Reluctance Stepper Motor**
3. **Hybrid Stepper Motor**

Advantages

- High positioning accuracy
- Good speed control
- No feedback system required
- Reliable operation

Disadvantages

- Lower efficiency

- Can lose steps under heavy load
- Generates vibration

Applications in Robotics

- CNC machines
 - 3D printers
 - Robotic arms
 - Precise positioning systems
-

3. Servo Motor in Robotics

Definition

A **servo motor** is a **closed-loop motor system** used for **precise control of position, speed, and torque**.

Components

A servo motor system includes:

- **DC motor**
- **Gear system**
- **Position sensor (encoder/potentiometer)**
- **Control circuit**

Working Principle

Servo motors operate using **feedback control**.

1. The controller sends a signal to the servo motor.
2. The motor moves to the desired position.
3. The sensor measures the position.
4. Feedback is sent to the controller.
5. If there is an error, the motor adjusts until the correct position is reached.

Types of Servo Motors

1. **DC Servo Motor**
2. **AC Servo Motor**

Advantages

- High precision control
- High torque
- Smooth movement
- Fast response

Disadvantages

- Higher cost
- Complex control system
- Requires feedback sensors

Applications in Robotics

- Robotic arms
- Humanoid robots
- Camera control systems
- Industrial automation

Comparison of DC Motor, Stepper Motor, and Servo Motor

Feature	DC Motor	Stepper Motor	Servo Motor
Motion Type	Continuous rotation	Step-by-step rotation	Controlled rotation
Control	Simple	Pulse control	Feedback control
Accuracy	Low	Medium	High
Cost	Low	Medium	High
Applications	Mobile robots	CNC machines	Robotic arms

Power Transmission in Robotics – Types of Gears

1.1 Introduction to Power Transmission

Power transmission in robots refers to the **method used to transfer motion and torque from motors to robot joints or moving parts**. Since motors rotate at high speed but low torque, gears are used to **reduce speed and increase torque**.

Functions of Gears

- Transmit mechanical power
 - Change speed and torque
 - Change direction of motion
 - Maintain synchronization between moving parts
-

1.2 Gear Terminology

Important terms used in gear systems:

Term	Description
Pitch Circle	Imaginary circle that rolls without slipping
Pitch Diameter	Diameter of pitch circle
Module	Ratio of pitch diameter to number of teeth
Gear Ratio	Ratio of number of teeth between gears
Backlash	Small clearance between mating teeth

Gear Ratio

$$\text{Gear Ratio} = \frac{\text{Number of teeth on driven gear}}{\text{Number of teeth on driving gear}}$$

1.3 Types of Gears

1.3.1 Spur Gear

Spur gears are the **most common type of gear** used in robotics.

Characteristics

- Teeth are **straight and parallel to shaft**
- Used for **parallel shafts**
- Simple design
- Efficient power transmission

Advantages

- Simple and inexpensive
- High efficiency ($\approx 98\%$)
- Easy to manufacture

Disadvantages

- Noisy at high speed
- Sudden tooth engagement causes vibration

Applications

- Robot joints
 - Gearboxes
 - Machine tools
-

1.3.2 Helical Gear

Helical gears have **teeth cut at an angle to the gear axis**.

Characteristics

- Teeth form a **helix shape**
- Smooth and quiet operation
- Higher load carrying capacity

Advantages

- Smooth motion
- Less noise
- Can transmit more power than spur gears

Disadvantages

- More expensive
- Generates axial thrust

Applications

- Industrial robots
 - Automotive transmission
 - High-speed machinery
-

1.3.3 Bevel Gear

Bevel gears are used to **transmit motion between intersecting shafts**.

Characteristics

- Cone-shaped gears
- Usually operate at **90° angle**

Types

- Straight bevel gear
- Spiral bevel gear
- Hypoid gear

Advantages

- Change direction of power transmission
- Efficient motion transfer

Applications

- Robot arm joints
- Differential drives
- Steering mechanisms

1.3.4 Worm Gear

A worm gear consists of a **worm (screw) and a worm wheel**.

Characteristics

- Shafts are **perpendicular**
- Provides **very high gear reduction**

Advantages

- High torque output
- Self-locking capability
- Compact design

Disadvantages

- Lower efficiency due to friction
- Generates heat

Applications

- Robotic manipulators
- Lifting mechanisms
- Conveyor systems

1.3.5 Rack and Pinion Gear

This mechanism converts **rotational motion into linear motion**.

Components

- Rack (straight gear)
- Pinion (circular gear)

Advantages

- Simple mechanism
- Accurate linear movement

Applications

- Robot linear actuators
 - CNC machines
 - Steering systems
-

1.3.6 Planetary Gear System

Planetary gears consist of:

- **Sun gear**
- **Planet gears**
- **Ring gear**

Characteristics

- Compact design
- High torque transmission
- Multiple gear ratios

Advantages

- High power density
- Smooth power distribution
- Efficient torque multiplication

Applications

- Robot gearboxes
 - Industrial automation
 - Precision robotic arms
-

2. Robotic Sensors

2.1 Introduction

Sensors are devices that **detect physical parameters from the environment and convert them into electrical signals.**

Robotic sensors help robots:

- Perceive the environment
- Detect obstacles
- Measure position and motion
- Control operations

Sensors act as the **sensory system of a robot.**

2.2 Classification of Robotic Sensors

Robotic sensors are mainly classified into two categories:

1. Internal Sensors (Proprioceptive Sensors)

Measure the **internal state of the robot.**

Examples:

- Position
- Velocity
- Acceleration
- Joint angle

Examples of internal sensors:

- Encoders
- Gyroscopes
- Accelerometers

2. External Sensors (Exteroceptive Sensors)

Measure **external environment information.**

Examples:

- Distance
- Temperature
- Light
- Sound

Examples:

- Camera
- Ultrasonic sensor

- Infrared sensor
-

2.3 Types of Robotic Sensors

2.3.1 Position Sensors

These sensors measure the **position of robot joints or links**.

Types

- Potentiometer
- Optical encoder
- Linear displacement sensor

Applications

- Joint angle measurement
 - Robot arm positioning
-

2.3.2 Velocity Sensors

Measure **speed of robot movement**.

Examples:

- Tachometers
- Optical encoders

Applications:

- Motor speed control
 - Motion monitoring
-

2.3.3 Force and Torque Sensors

Measure **force applied by robot arms**.

Applications

- Assembly operations
 - Gripping delicate objects
 - Force control
-

2.3.4 Proximity Sensors

Detect **presence of nearby objects without contact**.

Types

- Inductive sensors
- Capacitive sensors
- Ultrasonic sensors
- Infrared sensors

Applications

- Obstacle detection
 - Object detection
 - Industrial automation
-

2.3.5 Vision Sensors (Camera Systems)

Vision sensors help robots **see and interpret visual information**.

Components

- Camera
- Image processor
- Computer vision algorithms

Applications

- Object recognition
 - Quality inspection
 - Autonomous navigation
-

2.3.6 Tactile Sensors

These sensors detect **touch, pressure, and texture**.

Characteristics

- Mimic human skin
- Provide feedback during grasping

Applications

- Robotic grippers
- Medical robots
- Humanoid robots

2.3.7 Range Sensors

Measure **distance between robot and objects**.

Examples:

- Ultrasonic sensors
- LiDAR sensors
- Laser range sensors

Applications

- Autonomous vehicles
- Mobile robots
- Mapping and navigation

2.4 Sensor Characteristics

Important parameters of sensors:

Parameter	Description
Accuracy	Closeness to actual value
Resolution	Smallest measurable change
Sensitivity	Ability to detect small changes
Response Time	Time taken to produce output
Range	Minimum to maximum measurable value

2.5 Applications of Robotic Sensors

Robotic sensors are used in:

- Industrial automation
- Autonomous vehicles
- Medical robots
- Space exploration
- Manufacturing systems
- Service robots